Defence presentation outline

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1. Content (1 min)
   (a) I'll walk you through back ground my thesis addresses (Intro), both of these chapters (1 and 2) will address this question, but I'll start with the background first and finish with a conclusion; will last 15-20 minutes.
   (b) We don't have a lot of info on growing season length and growth relationship; I'll add data to this question in two systems that I find interesting.
2. Introduction
   (a) ACC lengthens growing seasons
      i. Happens through earlier bigger shifts in SOS, and smaller delayed EOS
      ii. (Figure): conceptual figure simplified showing SOS and EOS shifts
   (b) GS are longer, but also warmer which should increase growth
      i. Total annual GDD increases
      ii. (Figure): conceptual figure with GDD accumulation curve
   (c) SOS + EOS shifts = more GDD, but shifts are assymetric (both in how many days, but also for warmth)
      i. Impacts how much GDD is accumulated in the GS
      ii. SOS shift more, but are cooler
      iii. EOS shift less, but are warmer
      iv. (Figure): conceptual figure with showing how much GDD in each season
   (d) Recent studies failed to find this relationship (growth X GSL and thermal season)
      i. Important because temperate forest= 40% of global net carbon sink
      ii. (Figure): TBD
   (e) Question: do trees grow more with longer and warmer seasons?
      i. Objectives: address GS/growth relationship using two systems
      ii. Chapter 1: juvenile trees, 4 species, 4 prov, leaf phenology and tree rings
      iii. Chapter 2: mature trees, 11 species, leaf phenology and tree rings
      iv. (Figure): simplified version of thesis overview figure
3. Chapter 1
   (a) Introduction:
      i. Drought and competition could constrain growth
      ii. Internal constraints (local adaptation?)
      iii. Answer if juvenile trees grow more in a drought and competition limited environment
      iv. (Figure):
   (b) Methods: common garden set up
      i. Trees grown from seeds, 4 species, 4 provenances
      ii. Phenology for 3 years
      iii. Calendar and thermal season from leafout to budset
      iv. Growth via tree rings
      v. Bayes hierarchical model

        vi. (Figure): map and common garden photo

    (c) R/D: thermal > calendar (question reminder)

        i. Calendar season increased growth only for 2 species. Weaker effect is what people found in the litterature
        ii. 3 species grew more with warmer thermal season
        iii. Species grew more with warmer seasons than longer calendar season
        iv. Because thermal is a better predictor of growth
        v. Thermal season weights growth days while calendar assumes a stable growth rate throughout the season
        vi. (Figure): 2 pannel mu plot for GDD and GSL

    (d) R/D: EOS > SOS;

        i. EOS temperature higher temperature than SOS
        ii. EOS delayed less, but contribution is higher because days are warmer
        iii. Suggests looking more at EOS than SOS
        iv. (Figure): 2 pannel mu plot for EOS and SOS AND conceptual figure showing GDD differences at EOS and SOS

    (e) R/D: No provenance effect;

        i. In this small climatic gradient and range, no evidence of local adaptation on growth
        ii. Transition from juvenile to mature trees
        iii. (Figure): provenance mu plot

4. Chapter 2

    (a) Introduction: (we have more data (but same data of rings and pheno), so well be able to ask this additional question on previous year conditions with a perpective on life history strategies could inform tree growth responses

        i. Fast vs slower growing species
        ii. Diffuse vs ring porous species
        iii. Carbon allocation to storage = blur annual growth response to thermal and calendar seasons

    (b) Methods: citizen science leaf phenology at the Arboretum

        i. Growth via tree rings
        ii. Leaf phenology
        iii. 11 angiosperm deciduous species spanning different growth strategies and wood porousness
        iv. Bayes hierarchical model
        v. (Figure): Arboretum map AND core scans

    (c) R/D: weak response across species across current year predictors

        i. Ball decreased growth: southernmost range reduction in future ACC
        ii. Tiam: grew less with earlier SOS despite being fast growing
        iii. No consistent trend across life history strategies or wood porousness
        iv. (Figure): mu plots 4 facet

    (d) R/D: 2 species responded to previous year thermal season

        i. Ball increase growth with previous yr condition
        ii. Bnig grew more only when considering previous yr condition
        iii. Maybe suggests importance of carbon allocation to storage for mature trees; blur responses
        iv. (Figure): previous year conditions mu plot

5. Conclusion and future work

    (a) Chapter 1 and 2

        i. Chapter 1: warmer seasons for juvenile trees = good for forest regeneration and range expansion
        ii. Chapter 2: no effect of longer and warmer seasons: probably bad for forest carbon
        iii. (Figure): simplified version of thesis overview figure